My Idea:
As the objective is to find a quantifiable metric for what each position in an existing dense document embedding means, one of the ways we can do this is by deploying a multi-label neural network. I took inspiration from the GRaCE approach.
The variables of note are **d:** positions in the embedding set by the embedding model, **D:** concepts you ask about, set either by us our our algorithm, **N:** documents in the dataset, **Labels:** a yes/no for each document and concept (N x D), **Weights:** one row of d weights per concept (so a weight matrix of size D x d). Training is what produces these weights, but the interpretation at the end is done by us.
Step-by-step:

1. We'd first choose the anchors, which GRaCE picks algorithmically to cover the concept space and avoid near–duplicates simultaneously. I haven't read GRaCE in depth so I can't confirm exactly how it works, but I'm familiar with the approach.
2. We'd then collect many varied documents, preferably hundreds to thousands, including ones that don't contain each concept. A single document can't work for the express reason that meaning only shows up through context. A document that mentions the word "Japan" and a document that is about Japan are two completely different things.
3. Each document should be labeled yes/no per anchor with an independent labeler (an LLM or human). We'd have one written rule for what "yes" means. The labeler is separate from the embedding model, which avoids circularity. To put it in perspective, the LLM labels a document about cars in early 20th Century Japan as "yes" for "Cars," "Japan," and "20th Century History." If the document doesn't mention "Lions," it'll have "no" for "Lions."
4. Each document is embedded with a single model, same settings throughout, ensuring every document has a standardized positional embedding across the dataset.
5. Train a single layer: one sigmoid neuron per anchor, each with d weights, on the embedding → yes/no. Hold out about 20% of the documents for testing. What we want is to fine tune a document for which we know the labels (supervised learning) until the network produces the expected output of labels (multi-label essentially).
6. We'd then want to check the accuracy. If it's a near chance, the weights mean nothing. Using accuracy alone may not be useful as a lot of the labels are likely to be "no," so we might instead use F1 or precision/recall.
7. Read the weights. The highest-magnitude weights for each anchor/concept's row are what indicate the candidate positions. This is the hypothesis we make. So for instance if positions 46 and 48 have the highest weights across the Japan concept row, there's strong evidence those positions carry information that predicts Japan. To explain this, just consider the weight matrix (D x d), where positions 46 and 48 have the highest weights in the row for Japan (from 1…D). This concept row is trained across multiple documents, and usually, information about Japan is captured in those positions, though this is a hypothesis we test in the next step, not something the weights alone establish. How granular or specific we can get (instead of just Japan, say Japanese culture, Japanese economics, Japan's legal system, etc), depends on how many anchors we choose to create, on whether we have enough positive and hard-negative documents for each one, and on the embedding model.
8. Confirm Hypothesis: Once the highest-weighted positions for each concept have identified that concept's candidate positions, we can ablate them and reconstruct the source document to see how much information was captured in those positions, by comparing the ablated reconstruction

against the original document. This could prove useful for document loss identification. Ablation just means that we zero out the positions with the highest weights, say the top 10, then 50, 100, etc., for each concept, and watch both: A. how the probe's accuracy (or F1) falls, and B. how much of the source document is lost after reconstruction. As a control, we repeat both measurements after zeroing the same number of randomly chosen positions.

With this method we can not only find per-position scores for each anchor, we can also uncover how concentrated a concept is in certain positions based on how much accuracy falls and how much reconstruction degrades when those positions are ablated, compared to ablating the same number of random positions. The caveats of course are that weights show positions that carry predictive information regarding a concept, but that doesn't mean the embedding model actually uses those positions to represent it. Weights can split between overlapping positions, and concepts are probably spread across many positions, not one each. Results apply only to the embedding model , documents, and labeling rule that we test, and only concepts on the list are explained. However, the methodology behind this system is built to be applied to any other set of documents, anchors, and even embedding and labeling models, as long as we can get the raw embedding vectors.

Related work for the "how we differ" paragraph

- **Research I have planned:** Probing (Belinkov), plus V-information (Xu et al.) and MDL probing (Voita & Titov) as ways to score it → https://arxiv.org/pdf/2102.12452
- SAE-on-embeddings (O'Neill et al., Ye et al., Jiang et al.): finds features and labels them afterward, with no per-position score.
- RaDE/GRaCE: meaning built in by design, for new embeddings only.
- Sparse probing is the area to search before claiming novelty. I haven't verified anything there.

Open questions for the team

- Which embedding model. I'd personally suggest OpenAI's text-embedding-3-small or text-embedding-3-large hosted on Azure. Does it have a length limit?
- Which labeler, and how will we validate it?
- Does GRaCE's anchor closeness depend on a specific embedding model?

Controls:

Random-position ablation (same count as the top-weight positions, on both F1 and reconstruction).

- Checks whether zeroing out the concept positions hurts more than zeroing out any positions randomly.

Held-out 20% test set

The baseline for reconstruction is the unablated embedding's reconstruction. This is a baseline for reconstruction quality.

Specificity check (does ablating positions for one suspected concept interfere with the others).

Our baseline for performance is bag-of-words: can we predict each concept's labels with comparable accuracy from the presence of words alone, rather than from positions in the text embedding? This is a baseline for probe performance (F1).

Example:

Doc 1 (Cars in Japan):

Concepts: Cars, Japan, Trees.

[yes, yes, no]. → Cars and Japan → Doc.

Doc 1 —> embedding [x, y, z, a].
W1, w2, w3, w4 —>

Embedding [**w1x**, w2y, w3z, w4a]
W2 = w3 = w4 = 0
W1 >>>>.